BRAND-CONSTRAINED LATENT DIFFUSION FOR BROADCAST PACKAGING

A controllable background generator paired with an editable foreground compositor

learned constraints

Design brief

tone · palette · channel
programme intent · duration

Brand LoRA (rank 4)

learns brand appearance
without moving the base model

Brand kit

logo · type system
colour tokens · key art

Layout controller

Canny condition preserves
composition and edge rhythm

Spatial / motion plan

Canny edges · safe zone
trajectory: x, y, scale, rotation

Temporal residual

timestep-conditioned Δz
applied only in late denoising

FROZEN LATENT DIFFUSION

noise ε_t + prompt

z_t

z_(t-1)

late gate
t ≤ 0.30

denoise → decode

frozen backbone

editable output layer

Editable compositor

generated background
+ deterministic logo / type
+ safe-zone / trajectory
+ alpha compositing

Sequence delivery

16-frame ident / endboard
GIF + per-frame PNG
source noise retained
for reproducibility

RENDERED OUTPUT

Fixed held-out sample
commons_0150_v00 · seed 20262903
complete late-gated pipeline

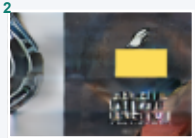
qualitative inspection only; statistical inference is in Fig. 2



frame 00



frame 08



frame 15